

Bags, Vectors & Transformers

Introduction · Day 1, Part 1

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Today's Session

Welcome & Introductions

Who we are — and who you are

What is Computational Text Analysis?

The big picture

Why Should You Care?

Research examples and possibilities

What Could This Mean for Your Research?

Your data, your questions

Course Overview

The 4-day journey ahead

Practicalities

Tools, schedule, and expectations

Meet Your Instructor



Denise J. Roth

PhD Candidate · Strategic Communication Group
Wageningen University & Research

- ✍ Online political communication — incivility, polarization, public discourse
- 🔍 Black digital health communication and scientization of public debate
- 🧠 Computational text analysis as a method across all of the above

Feel free to ask questions throughout — this is a participatory course!

Now You! 🖐️

Take a moment to introduce yourself to the group:

 Your name and department

 What you are researching

 What kind of text data you work with (or wish you could)

 Prior experience with methods or software (R, Python, SPSS... anything!)

 One thing you hope to get out of this course

What is Computational Text Analysis?

Using computers to systematically extract patterns and meaning from large collections of text — at a scale no human reader could manage alone.

What kind of questions?

- How does framing of topic X change over time?
- What themes dominate in corpus Y?
- How similar are these two sets of documents?
- Does sentiment differ across groups?

What kinds of text?

- News articles, social media posts
- Parliamentary speeches, policy documents
- Interview transcripts, open survey responses
- Pretty much anything in written form

The method is flexible — what matters is having a research question text can answer.

Quantitative or Qualitative?

Spoiler: it's both — and that's what makes it powerful.

The execution is quantitative

- Counts, frequencies, scores
- Algorithms and models
- Large N , reproducible
- Systematic and transparent

The interpretation is qualitative

- You still assign meaning
- You validate by reading
- Context always matters
- Numbers don't speak for themselves

Meaning-making is always qualitative

- Counting words $\neq$ understanding text
- CTA surfaces patterns
- You interpret what they mean
- The researcher is never out of the loop

Think of CTA as a tool, not a paradigm — it serves your research question, whatever that is.

Why Should You Care?

 Analyzing 100,000 news articles to track how migration framing shifts over time

 Classifying sentiment in millions of tweets after a political crisis

 Comparing parliamentary speeches across 20 years and 5 countries

 Making sense of thousands of open-ended survey responses without losing the nuance

What Could This Mean for Your Research?

Think about your own work for a moment. . .

 What text data do you already have, or could realistically collect?

 What research question could you explore if you could read 10,000 texts?

 Where does analysis currently take the most time — and could it be automated?

We'll come back to your answers throughout the week!

The Journey Ahead — 4 Days

DAY 1

Text Data in Python

- Python recap
- Loading & exploring text
- Tokenization

DAY 2

Bag-of-Words

- Word frequencies
- TF-IDF
- Topic modeling (LDA)

DAY 3

Static Embeddings

- Word2Vec & GloVe
- Pre-trained models
- Similarity & viz.

DAY 4

Contextual Embeddings

- Transformers & BERT
- HuggingFace
- Your own data

Practicalities



Schedule

10:00 – 12:30	Morning
12:30 – 13:30	Lunch
13:30 – 16:00	Afternoon



Tools

Google Colab (browser-based,
no installation needed!)

NLTK, scikit-learn,
gensim, transformers

Links to notebooks provided



Prerequisites

Basic Python
(DataCamp level)

No prior CTA knowledge
needed — we start from
scratch

Let's get started!

Any questions before we dive into Python and text data?